TRIPATHI STUDY ZONE
JEE & NEET
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CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 3

(With Solutions)

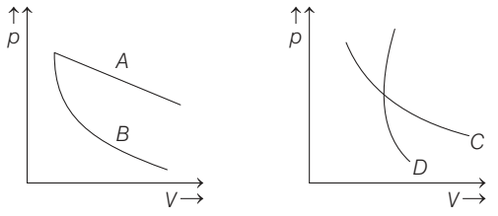
Duration : 3 Hours **Max. Mark : 720**

General Instructions

- The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- The question paper contains three parts of Physics, Chemistry and Biology respectively.
- Part A contains 45 Questions from Physics section. Part B contains 45 Questions from Chemistry section. Part C contains 90 Questions from Biology.

PART A PHYSICS

1. In the following figure, four curves A, B, C and D are shown below. The curves are



(a) A and C for isothermal, while B and D for adiabatic
(b) A and C for adiabatic, while B and D for isothermal
(c) A and B for isothermal, while C and D for adiabatic
(d) A and B for adiabatic, while C and D for isothermal

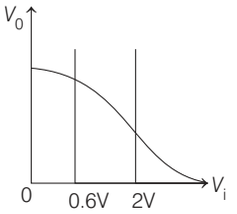
2. A particle is executing SHM of amplitude 8 cm. At what displacement from the mean position, the energy is half kinetic and half potential?
(a) 4 cm (b) $2\sqrt{2}$ cm
(c) $4\sqrt{2}$ cm (d) $8\sqrt{2}$ cm

3. Three particles each of mass m are placed at the corners of an equilateral triangle of side d. If the side of the triangle is changed from d to $2d$, then work done on this system is
(a) $-\frac{3Gm^2}{2d}$ (b) $\frac{3Gm^2}{2d}$ (c) $\frac{Gm^2}{2d}$ (d) $-\frac{3Gm^2}{d}$

4. Two circular loops P and Q are made of same wire and their radii are in ratio $1:n$. Their moment of inertia about the axis passing through the centre and perpendicular to their plane are in ratio $1:m$. Then, m and n are related as
(a) $m = n$ (b) $m^2 = n$
(c) $m = n^2$ (d) $m = n^3$

5. A step-up transformer of turn ratio of 10 has 500 turns in primary winding. The primary is connected to an AC supply 20 V, 50 Hz. The secondary will have an output of
(a) 200 V, 50 Hz (b) 400 V, 100 Hz
(c) 100 V, 50 Hz (d) 150 V, 100 Hz

6. The figure below shows, the transfer characteristics of a base biased CE transistor. Which of the following statment is true?



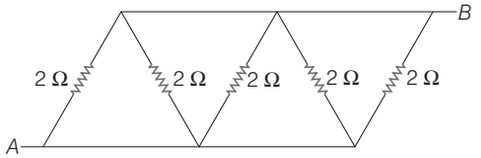
(a) At V_i = 0.4 V, transistor is in active state
(b) At V_i = 1V, it can be used as an amplifier
(c) At V_i = 0.5V, it can be used as a switch truned ON
(d) At V_i = 2.5 V, it can be used as a switch truned OFF

7. A wooden block floats in water, where 50% of its volume is outside water. When the same block floats in alcohol 70% of its volume remains outside the alcohol. The relative density of alcohol is
(a) 1.78 (b) 1.50 (c) 1.66 (d) 2.1

8. A disc of mass 40g is kept floating horizontally by throwing 20 marbles per second against it from below. If the mass of each marble is 4.9 g, then the velocity with which the marbles are striking the disc is
(assume that, the marbles strike normally and rebound downward with same speed)
(a) 9.8 ms^{-1} (b) 4.9 ms^{-1} (c) 5 ms^{-1} (d) 2 ms^{-1}

9. In photo electric emission from a metallic surface, wavelength of incident light is 2×10^{-7} m and stopping potential is 2.5V. The threshold frequency of the metal in Hz approximately
(charge on electron, $e = 1.6 \times 10^{-19}$ C.
Planck's constant, $h = 6.6 \times 10^{-34}$ J-s)
(a) 12×10^{16} (b) 9×10^{15} (c) 9×10^{14} (d) 9×10^{16}

10. In the diagram shown below, equivalent resistance between A and B is



(a) 7.5 Ω (b) 5 Ω
(c) 2.5 Ω (d) 10 Ω

11. A body when projected from Earth surface in vertical direction, attains a height of 18 m. If it is projected with the same velocity upwards from a planet having density 3 times and radius half of Earth, then will rise to
(a) 12 m (b) 10 m
(c) 15 m (d) 20 m

12. Light from two coherent sources of the same amplitude A and wavelength λ illuminate the screen. The intensity of the central maximum is I_0. If the sources are incoherent, the intensity at the same point will be
(a) $4I_0$ (b) $2I_0$ (c) I_0 (d) $\frac{I_0}{2}$

13. A body A is thrown up vertically from the ground with a velocity v_0 and another body B is simultaneously dropped from a height H. They meet at a height $\frac{H}{2}$, if v_0 is equal to
(a) $\sqrt{2gH}$ (b) $\sqrt{gH}$ (c) $\frac{1}{2}\sqrt{gH}$ (d) $\sqrt{\frac{2g}{H}}$

14. The forbidden energy gap in Ge is 0.75 eV, given hc = 11250 eVÅ. The maximum wavelength of radiation that will generate electron hole pair is
(a) 1500Å (b) 15000Å (c) 150.52Å (d) 150Å

15. Activity of a radioactive sample decreases to (1/3)rd of its original value in 3 days. Then, in 9 days, its activity will become
(a) (1/27) of the original value (b) (1/9) of the original value
(c) (1/18) of the original value (d) (1/3) of the original value

16. A charged particle q is shot with speed v towards another fixed charged particle Q. It approaches Q upto a closest distance r and then returns. If q were given a speed $2v$. The closest distance of approach would be



(a) r (b) $2r$ (c) $r/2$ (d) $r/4$

17. A test tube weighing 10 g and external diameter 2 cm is floated vertically in water by placing 10 g of mercury at its bottom. The tube is depressed in water a little and then released. Find the time of oscillation. (Take, $g = 10\ ms^{-2}$)
(a) 0.5 s (b) 1 s (c) 5 s (d) 3 s

18. Two rods of copper and aluminium of length L_1 and L_2, respectively. The coefficient of linear expansion of both metal rod are α_1 and α_2, respectively. If they are joined together and $(L_1 + L_2)$ is maintained same at all temperatures, which one of the following relations holds good?
(a) $\alpha_1 L_2^2 = \alpha_2 L_1^2$ (b) $\alpha_1^2 L_2 = \alpha_2^2 L_1$
(c) $-\alpha_1 L_1 = \alpha_2 L_2$ (d) $\alpha_1 L_2 = \alpha_2 L_1$

19. A body of mass 4 kg is acted upon by a force which causes a displacement in it given by $x = t^2$ m, where t is time in second. The work done by force in 4s is
(a) 64 J (b) 128 J (c) 200 J (d) 240 J

20. Three objects coloured, black, red and white can withstand hostile condition at 3200° C. These objects are thrown into furnace, where each of them attains a temperature of 2335° C. Which object will glow brightest?

(a) Black (b) Red
(c) White (d) All equally glow

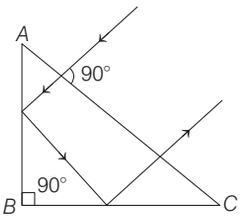
21. If a radioactive nucleus decays by two different processes, the half-life for first process is t_1 and for second process it is t_2. Then, the effective half life t of the nucleus is given by

(a) $t = t_1 + t_2$ (b) $\sqrt{t} = \sqrt{t_1} + \sqrt{t_2}$
(c) $t^{-1} = t_1^{-1} + t_2^{-1}$ (d) None of these

22. A light emitting diode(LED) has a voltage drop of 2V across it and passes a current of 10 mA. When it operates with a 6V battery through a limiting resistor R, the value of R is

(a) 40 KΩ (b) 4 KΩ (c) 200Ω (d) 400Ω

23. A ray falls on a prism $ABC(AB = BC)$ and travels as shown in the figure. The minimum refractive index of the prism material should be



(a) 4/3 (b) $\sqrt{2}$ (c) 1.5 (d) $\sqrt{3}$

24. Work done by 3 mole of gas at 47° C to triple its volume at constant pressure is
(R = 2 cal mol^{-1} °C^{-1})

(a) 3402 cal (b) 3428 cal
(c) 3832 cal (d) 3840 cal

25. Escape velocity of a body from Earth is 11 km s^{-1}. Assuming the mass and radius of Earth to be about 81 times and 4 times the mass and radius of the Moon, the escape velocity (in km/s) from the surface of the Moon will be

(a) 0.54 (b) 2.44 (c) 11 (d) 22

26. A current of I ampere flows along infinitely long conductor, then magnitude of magnetic field at any point on the conductor is

(a) infinity (b) zero
(c) $\frac{\mu_0 I}{4\pi r}$ (d) $\frac{\mu_0 I}{2\pi r}$

27. If an X-rays tube operates at 18 kV, then the maximum speed of electron striking the anode is

(a) 2×10^7 m/s (b) 3×10^8 m/s
(c) 8×10^7 m/s (d) 8×10^{10} m/s

28. Object distance $u = (50.1 \pm 0.5)$ cm and image distance $v = (20.1 \pm 0.2)$ cm, then focal length is

$\left[\frac{1}{f} = \frac{1}{u} + \frac{1}{v}, \text{where } u \text{ and } v \text{ are} + ve\right]$

(a) (12.4 ± 0.4) cm (b) (12.4 ± 0.1) cm
(c) (14.3 ± 0.4) cm (d) (14.3 ± 0.1) cm

29. If a particle of mass m and charge q is thrown at a speed u against an uniform electric field E. The distance that it will travel before coming to momentary rest is

(a) $\frac{mu}{qE}$ (b) $\frac{qE}{m}$
(c) $\frac{mu^2}{2qE}$ (d) None of these

30. A projectile is projected at an angle of 45° with the horizontal at a speed of 12 ms^{-1}. A second object is simultaneously released from a point on the vertical line along maximum height of first projectile, where they collide. The inital height of second object is (g = 10 ms^2)

(a) 3.6 m (b) 7.2 m (c) 2.5 m (d) 5 m

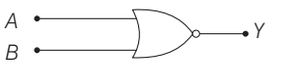
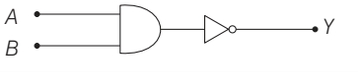
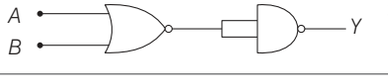
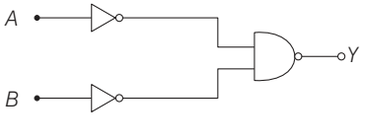
31. Magnetic field of the earth is H = 3 g. A magnet is vibrating 5 oscillations per minute, then the reduction required in the magnetic field of the earth to increase time period upto 10 oscillations per minute is

(a) 2.25 g (b) 0.6 g (c) 0.9 g (d) 0.12 g

32. A circular disc rolls down an inclined plane without slipping. What fraction of its total energy is translational?

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{2}{3}$ (d) $\frac{1}{\sqrt{3}}$

33. Match the symbol gates or gate circuits given in Column I to the gate given in Column II and choose the correct option from the codes given below.

Column I		Column II
A.		1. NOT
B.		2. OR
C.		3. NOR
D.		4. NAND
		5. AND

Codes

	A	B	C	D		A	B	C	D
(a)	3	2	4	5	(b)	3	4	1	2
(c)	5	4	3	2	(d)	5	4	1	2

34. Electron in hydrogen atom first jumps from third excited state to second excited state and then from second excited state to first excited state, then the ratio of wavelengths $\lambda_1 : \lambda_2$ emitted in two cases is

(a) 27 : 5 (b) 7 : 5 (c) 20 : 7 (d) 27 : 20

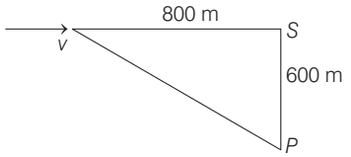
35. A bat moving at 10 ms^{-1} towards a wall sends a sound signal of 8000 Hz towards it. On reflection, it hears a sound of frequency f. The value of f in Hz is close to (speed of sound = 320 ms^{-1})

(a) 9500 Hz (b) 8516 Hz (c) 8000 Hz (d) 8425 Hz

36. In an ammeter, 0.4% of main current passes through galvanometer. If resistance of galvanometer is R_g, then shunt resistance of ammeter will be

(a) $\frac{R_g}{299}$ (b) $\frac{R_g}{120}$ (c) $\frac{R_g}{200}$ (d) $\frac{R_g}{249}$

37. A person P is 600 m away from the station. When, train is approaching station with 72 km h^{-1}, it blows a whistle of frequency 800 Hz, when 800 m away from the station. If speed of sound is 340 ms^{-1}, then the frequency heard by the person is

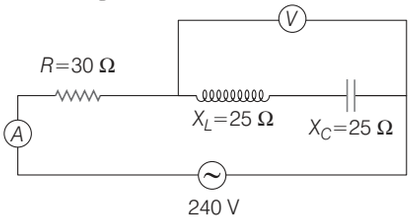


(a) 800 Hz (b) 839.5 Hz (c) 829.5 Hz (d) 843.5 Hz

38. A capacitor is charged in such a manner that it stores energy of 2 J. Now, potential energy is increased to 8 J, it is done by

(a) double the potential difference across the capacitor without changing the charge
(b) double both the potential difference and charge
(c) double the charge without changing the potential difference
(d) quadruple the potential difference across the capacitor without changing the charge

39. In the given circuit (as shown in the figure) neglecting source resistance, then voltmeter and ammeter reading will be



(a) 120 V, 3A (b) 120 V, 6A (c) 0V, 8A (d) 0V, 3A

40. In a radiation laboratory, a experiment is conducted with two spherical bodies X (radius 12 cm) and Y (radius 36 cm), they have temperature T_1 and T_2, respectively.

The maximum intensity in the emission spectrum of X is at 500 nm and in that Y is at 1500 nm. Considering them to be black bodies, what will be the ratio of the rate of total energy radiated by X to that of Y ?

(a) 9.5 (b) 9
(c) 8.5 (d) 8

41. In Young's double slit experiment, the 5th bright fringe from central maxima due to wavelength λ_1 coincides with 8th bright fringe from central maxima due to wavelength λ_2, then possible wavelengths of visible lights are

(a) 500 nm and 800 nm
(b) 480 nm and 400 nm
(c) 640 nm and 400 nm
(d) 640 nm and 560 nm

42. A copper wire of length 2m and radius 0.1 cm is joined in series with an iron wire of length 4 m and radius 0.4 cm and a current is passed through the wires. The ratio of the current density in iron and copper wires is

(a) 1 : 2 (b) 4 : 1
(c) 1 : 16 (d) 16 : 1

43. Three lenses of focal length +15 cm, + 150 cm and +250 cm are available, for making an astronomical telescope. To produced the largest magnification, the focal length of the eye-piece should be

(a) +15 cm
(c) +150 cm
(d) +250 cm
(d) None of these

44. The mass and density of the Moon, if acceleration due to gravity on its surface is 1.62 ms^{-2} and its radius is 1.74×10^6 m, are respectively (Take, $G = 6.67 \times 10^{-11} N - m^2 kg^{-2}$)

(a) 7.35×10^{22} kg, 3.3×10^3 kg m^{-3}
(b) 3.3×10^3 kg, 7.35×10^{22} kg m^{-3}
(c) 7×10^{22} kg, 4×10^3 kg m^{-3}
(d) 5×10^{22} kg, 3×10^3 kg m^{-3}

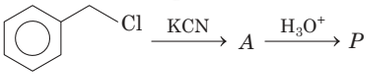
45. Electron revolving in circular path of radius r with speed v produced magnetic filed B at centre. Then, correct relation among v, B and r is

(a) $B \propto \frac{1}{v^2} \propto r^2$ (b) $B \propto v^2 \propto \frac{1}{r^2}$
(c) $B \propto v \propto \frac{1}{r^2}$ (d) $B \propto \frac{1}{v} \propto \frac{1}{r^2}$

PART B CHEMISTRY

46. Which one of the following complex is an outer orbital complex?
(a) $[Fe(CN)_6]^{4-}$ (b) $[Mn(CN)_6]^{4-}$ (c) $[Co(NH_3)_6]^{3+}$ (d) $[CoF_6]^{3-}$

47. The product P in following reaction is



(a) 2-phenylpropanoic acid (b) benzoic acid
(c) 2-phenylethanoic acid (d) phenylmethanoic acid

48. Formalin is a colourless solution in water that is mostly used as a preservative. It is also used as a disinfectant and an antibacterial. It is
(a) a solution of 40% HCHO and 60% CH_3CHO
(b) 40% aqueous solution of HCHO
(c) solution of fructose
(d) None of the above

49. Among the given compounds, the compound which is oxidised to prepare methyl ethyl ketone is
(a) *tert*-butyl alcohol (b) butan-2-ol
(c) propan-2-ol (d) butanol

50. In $[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$ species, the hybridisation state of Ni atom in all the complexes are respectively
(a) sp^3, dsp^2, dsp^2 (b) sp^3, dsp^2, sp^3
(c) $sp^3, sp^3\ dsp^2$ (d) dsp^2, sp^3, sp^3

51. Methyl alcohol when reacted with carbon monoxide using cobalt or rhodium as catalyst, compound A is formed. On heating A with HI in the presence of red phosphorus as catalyst, B is formed.
The compound B is
(a) CH_3COOH (b) CH_3CHO (c) CH_3CH_2I (d) $CH_3 \cdot CH_3$

52. The complex $[Ru(H_2O)_6]^{2+}$ and $[RuCl_6]^{3-}$ have nearly the same Δ_o values because
(a) both complexes contain the same metal atom
(b) both the complexes have a symmetrical octahedral structure
(c) the presence of weaker field ligand in one is compensated by the presence of the metal in a higher oxidation state
(d) H_2O and Cl^- occupy adjacent positions in the spectrochemical series

53. Although +3 is the characterstic oxidation state for lanthanoid yet cerium also shows +4 oxidation state because
(a) it has tendency to attain noble gas configuration
(b) it has variable ionisation enthalpy
(c) it resembles Pb^{4+}
(d) it has tendency to attain f^7 configuration

54. Which one of the following octahedral complexes will not show geometrical isomerism?
(a) $[MA_4B_2]$ (b) $[MA_5B]$
(c) $[MA_2B_4]$ (d) $[MA_3B_3]$

55. On heating 3, 3-dimethyl-butan-2-ol with conc. H_2SO_4, the major product obtained is
(a) 2, 3-dimethylbutene (b) 3, 3-dimethylbutene
(c) 2, 3-dimethylbut-2-ene (d) 3, 3-dimethylbut-2-ene

56. Among the given lanthanoid ion, the one which is not paramagnetic in nature is
(a) Yb^{2+} (b) Sm^{2+}
(c) Eu^{2+} (d) Ce^{2+}

57. Which one of the following represents correct order of basic strength?
(a) $NH_2OH < NH_3 < N_2H_4$ (b) $NH_3 < N_2H_4 < NH_4OH$
(c) $NH_2OH < N_2H_4 < NH_3$ (d) $N_2H_4 < NH_2OH < NH_3$

58. Which of the following statements is not correct?
(a) Brass is an alloy of Cu and Sn
(b) Transition elements form alloy due to their similar radii
(c) Steel is also an alloy
(d) German silver is an alloy of Cu, Zn and Ni

59. Which one of the following oxides is most acidic in nature?
(a) Bi_2O_3 (b) Sb_2O_3
(c) As_2O_3 (d) P_2O_5

60. The radial function depends upon quantum number n and l whereas angular functions depend upon quantum numbers l and m, i.e. these are independent of n. The total wave function may therefore, be written as

$$\psi(r, \theta, \phi) = \underset{\text{(Radial part)}}{r_{nl}} + \underset{\text{(Angular part)}}{\theta_{l,m}\ \phi_n}$$

The number of radial node present in $4d$-orbital is
(a) 3 (b) 2 (c) 1 (d) 0

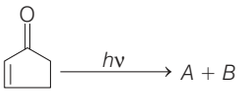
61. Among the following substituted silicone, the one which will give cross-linked silicone polymer on hydrolysis?
(a) R_3SiCl (b) R_4Si
(c) $RSiCl_3$ (d) R_2SiCl_2

62. The ions O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+} are isoelectronic. Their ionic radii show
(a) an increase from Na^+ to Al^{3+} and then decreases from O^{2-} to F^-
(b) a decrease from Na^+ to Al^{3+} and then increases from O^{2-} to F^-
(c) a significant increase from Al^{3+} to O^{2-}
(d) a significant decrease from Al^{3+} to O^{2-}

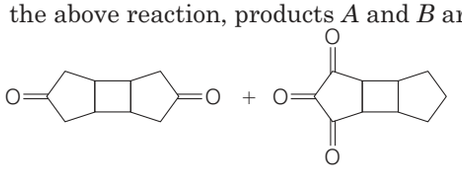
63. From the given sets of quantum number which is not consist with the theory?

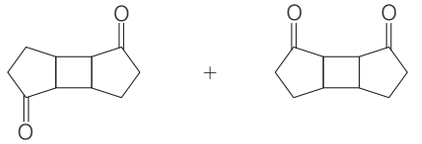
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(a)	1	1	1	$\frac{1}{2}$	(b)	3	0	0	$\frac{1}{2}$
(c)	1	0	0	$\frac{1}{2}$	(d)	2	0	0	$-\frac{1}{2}$

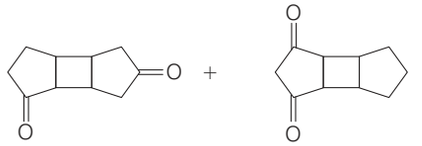
64.



In the above reaction, products A and B are

(a) 

(b) 

(c) 

(d) 

65. A, B and C are ideal gases. Their molecular weight are 2, 4 and 28 respectively. The rate of diffusion of these gases follow the order

(a) $C > A > B$ (b) $C > B > A$
(c) $A = B = C$ (d) $A > B > C$

66. Hess's law states "the enthalpy change accompanying a chemical reaction is independent of the route by which the chemical change occurs". Then, we can conclude that the Hess's law is used for determination of

(a) heat of reaction
(b) heat of formation
(c) heat of transformation
(d) All of the above

67. On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which of the following does not occur?

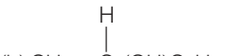
(a) Na^+ ions are formed in the solution
(b) Blue coloured solution is obtained
(c) Liquid NH_3 becomes good conductor of electricity
(d) Liquid NH_3 remains diamagnetic

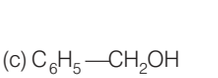
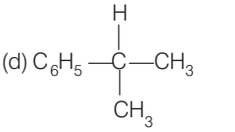
68. Which of the following is correct regarding following chemical reaction?

$$3H_2 + N_2 \rightleftharpoons 2NH_3$$

(a) $K_p > K_C$ (b) $K_p < K_C$
(c) $K_p = K_C$ (d) None of these

69. Which of the following is an optically active molecule?

(a)  (b) 

(c)  (d) 

70. 4 moles of X and 4 moles of Y are mixed together. For the reaction $X + Y \rightleftharpoons P + Q$, 2 moles of P and Q are formed at equilibrium. The equilibrium constant for the reaction will be

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 4

71. Which of the following statement is incorrect regarding boric acid?

(a) Boric acid has layered structure
(b) It is prepared by BCl_3 with H_2O
(c) It is used as mild antiseptic in washing eye
(d) It has square planar structure

72. From the following reactions,

$$R\overset{-}{O}\overset{+}{N}a + H_2O \rightleftharpoons ROH + NaOH$$

$$ROH + NaNH_2 \rightleftharpoons R\overset{-}{O}\overset{+}{N}a + NH_3$$

Which of the following order is correct regarding basic strength?

(a) $OH^- < NH_2^- < RO^-$ (b) $NH_2^- < OH^- < RO^-$
(c) $OH^- < RO^- < NH_2^-$ (d) $RO^- < NH_2^- < OH^-$

73. The entropy values (in $JK^{-1}\ mol^{-1}$) of $H_2 = 130.6$, $Cl_2(g) = 223.0$ and $HCl(g) = 186.7\ J\ mol^{-1}\ K^{-1}$ at 298 K and 1 atm pressure then entropy change for the reaction,

$$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g) \text{ is}$$

(a) +540.3 (b) −166.9
(c) +727.3 (d) +19.8

74. Which of the following statements is correct about resonance?

(a) Resonating structures are more stable than hybrid resonance
(b) Equivalent resonating structures do not contribute equally
(c) Resonance affects bond lengths but not bond angles
(d) Resonating structures indicate that the molecule has no real existence

75. $CH \equiv CH \xrightarrow{O_3/NaOH} X \xrightarrow{Zn/CH_3COOH} Y, Y$ is

(a) $\begin{matrix} CH_2OH \\ | \\ CH_2OH \end{matrix}$ (b) CH_3CH_2OH

(c) CH_3COOH (d) CH_3OH

76. It was found that a unique set of conditions was responsible for ozone hole. In summer season, NO_2 and CH_4 react with chlorine monoxide and chlorine atoms forming chlorine sinks, prevents much ozone depletion. Which of the following gas is harmful to ozone layer?

(a) CFC (b) MIC (c) Phosgene (d) CO_2

77. The C— C bond length of the following molecules is in the order

(a) $C_2H_6 > C_2H_4 > C_6H_6 > C_2H_2$
(b) $C_2H_2 < C_2H_4 < C_6H_6 < C_2H_6$
(c) $C_6H_6 > C_2H_2 > C_2H_6 > C_2H_4$
(d) $C_2H_4 > C_2H_6 > C_2H_2 > C_6H_6$

78. Diamond has a fcc structure with two atoms per lattice point and a unit cell of edge length 3.569 Å. What will be the density of diamond?

(a) 3.506 g cm^{-3} (b) 3 .0 g cm^{-3}
(c) 1.75 g cm^{-3} (d) 1 .5 g cm^{-3}

79. Normality and molarity changes with temperature because they involve volumes. The normality of 0.3 M phosphoric acid, H_3PO_4 is

(a) 0.9 N (b) 0.1 N
(c) 0.3 N (d) 0.6 N

80. The equivalent conductance of NaCl at concentration C and at infinite dilution are λ_C and λ_∞ respectively. The correct relationship between λ_C and λ_∞ is given as

(a) $\lambda_C = \lambda_\infty - B\sqrt{C}$ (b) $\lambda_C = \lambda_\infty - (B)C$
(c) $\lambda_C = \lambda_\infty + (B)C$ (d) $\lambda_C = \lambda_\infty + B\sqrt{C}$

81. For a reaction $4A \longrightarrow 5B$, if the rate of formation of B is y mol/L, then the rate of consumption of A will be

(a) $\frac{4}{5}y$ (b) $5y$ (c) y (d) $\frac{5}{4}y$

82. Molality of aqueous solution of 8.0 M ethanol having density 1.025 g/mL is

(a) 12.17m (b) 24.34m (c) 10.17m (d) 14.35m

83. Consider the following reactions,

$A \rightarrow X,\ k_A = 10^{18}\ e^{-8000/T}$

$B \rightarrow X,\ k_B = 10^{17}\ e^{-4000/T}$

Temperature (T/K) at which $k_A = k_B$ is

(a) 4000 K (b) 8000 K
(c) $\frac{4000}{2.303}$ K (d) $\frac{8000}{2.303}$ K

84. The process of gangue removal or increase in concentration of ore depends upon the nature of ore and the impurities present in it. Froth floatation process is an important method for the concentration of the ore. Which of the following ores involves the froth floatation method during extraction of metal?

(a) Chromite (b) Zincite (c) Bauxite (d) Iron pyrite

85. Most of the organic compounds are generally used in the formation of drugs. 1.1% solution of phenol is used as

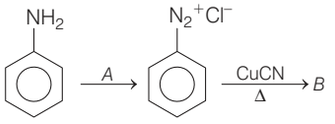
(a) antiseptic (b) antipyretic (c) disinfectant (d) tranquilizer

86. The rate of reaction $A \rightarrow B$ quadruples on increasing the concentration of A by 8 times, thus the order of the reaction is

(a) 4 (b) 1/2 (c) 2 (d) 1

87. Proteins when heated with conc. HNO_3 gives a yellow colour. This test used in identification of protein is known as

(a) xanthoprotic test (b) biuret's test
(c) acid base test (d) hoppe's test

88. 

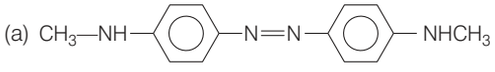
A and B respectively are

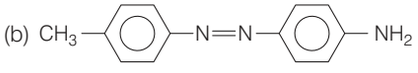
(a) phenol and methyl nitrile
(b) HNO_3 and benzonitrile
(c) $NaNO_2$/HCl and benzonitrile
(d) HNO_2/H^+ and methyl cyanide

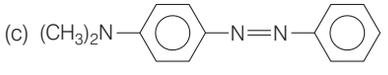
89. Which of the following method can be used to synthesise fluorobenzene in the laboratory?

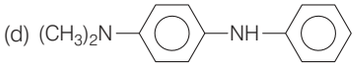
(a) Heating phenol with HF and KF
(b) Reacting bromobenzene with NaF solution
(c) Diazotisation of aniline followed by heating the diazonium salt with HBF_4
(d) Direct flourination of benzene with F_2 gas

90. Aniline when diazotised in cold and then treated with dimethylaniline gives a coloured compound, the product is

(a) 

(b) 

(c) 

(d) 

PART C BIOLOGY

91. The chloroplast thylakoids are in the form of
(a) interconnected tubules (b) interconnected sacs
(c) independent discs (d) stacked discs

92. The water column in xylem vessels or tracheids does not undergo rupture or fractionation during ascent of sap. This is possible due to
(a) lignified thick walls of vessels
(b) transpirational pull
(c) cohesion and adhesion forces
(d) imbibition pressure

93. Sclerenchymatous fibres are
(a) lignified (b) dead
(c) found in pericycle of stem (d) All of these

94. A pendulus spike bearing unisexual flowers is
(a) umbel (b) catkin
(c) spadix (d) corymb

95. Physiological barriers include
I. Tears from the eyes II. Saliva in the mouth
III. Acid in the stomach IV. Interferons
Choose the correct option.
(a) I and II (b) I, II and III
(c) II, III and IV (d) I and III

96. Pseudometamerism refers to
I. segmentation that simultaneously divides body both externally and internally.
II. the body is divisible into parts or segments called proglottid.
III. proglottids develop from the neck but are not embryonic in origin.
IV. repetition which appears due to the repeated budding.
Choose the correct option.
(a) I and II (b) II, III and IV
(c) III and IV (d) I, III and IV

97. The source of complementary RNA in RNA interference is
(a) infection by viruses having RNA genomes
(b) transposons that replicate *via* an RNA intermediate
(c) Both (a) and (b)
(d) None of the above

98. The action of an inhibitor which diminishes the catalysis *via* reverse binding at the active site of enzyme is best described as
(a) competitive (b) allosteric
(c) facultative (d) non-competitive

99. Which among the following statements are correct regarding ETC?
I. Cytochrome-a_3 donates the electrons to free molecular oxygen.
II. Oxygen acts as the last electron acceptor at the end of respiratory chain.
III. Oxygen also receives two protons directly from the aqueous medium to form one molecule of water.
IV. NADH and FADH bring electrons to the electron transport system.
Choose the correct option.
(a) I and II (b) II and III (c) III and IV (d) All of these

100. Which of the following is not an organ where lymphocytes originate, mature or multiply?
(a) Spleen (b) Bone marrow
(c) Thymus (d) Pancreas

101. Apospory is
(a) development of sporophyte without fusion of male and female gametes
(b) development of gametophyte from any cell of sporophyte other than spores
(c) Both (a) and (b)
(d) None of the above

102. Ommatidium is found in
(a) Porifera (b) Coelenterata (c) Annelida (d) Arthropoda

103. In humans, cartilage bone forms by the deposition of
(a) bony matter by osteoclasts
(b) resorption of bone matter by osteoblasts
(c) deposition of bony matter by osteoblasts and resorption by chondroclasts
(d) deposition of bony matter by chondrocytes

104. The common ancestor of man and ape is
(a) *Dryopithecus*
(b) *Australopithecus*
(c) *Hylobates hoolock* (gibbon)
(d) *Pongo pigmaeous* (orangutan)

105. What is a DNA molecule that is able to replicate autonomously called?
(a) Genome (b) Chromosome
(c) Replicon (d) Plasmid

106. A person with asthma has difficulty in breathing because of the constriction of
(a) bronchi (b) alveoli
(c) pharynx (d) terminal bronchioles

107. Filamentous thalli, hyphae generally non-septate and multinucleate are vegetative features of
(a) Zygomycetes – *Rhizopus*
(b) Basidiomycetes – *Chlamydomonas*
(c) Ascomycetes – *Neurospora*
(d) Deuteromycetes – *Mycorrhiza*

108. The flowers show trimerous symmetry in
(a) Dicotyledonae (b) Monocotyledonae
(c) Cryptogamae (d) Phanerogamae

109. 'Doom' is a mixture of
I. *Bacillus popilliae*
II. *Bacillus lentimorbus*
III. *Bacillus thuringiensis*
IV. *Hirrutella thompsoni*
Choose the correct option.
(a) I and III (b) I and II (c) III and IV (d) I and IV

110. The fallen stars is the name given to
(a) *Oscillatoria* (b) *Nostoc*
(c) *Zygnema* (d) *Cladophora*

111. Families in plant kingdoms are characterised based on which of the following features?
(a) Seasonal similarities and variations
(b) Anatomy of various parts
(c) External morphology
(d) Vegetative and reproductive parts

112. GIFT is a technique to overcome the problem of infertility in females who
(a) cannot produce an ovum
(b) cannot provide a suitable environment for fertilisation
(c) cannot maintain a viable foetus
(d) have two narrow cervical canal for sperm passage

113. Read the following statements.
I. Some species acquire heat from sun when the body temperature drops below the comfort zone.
II. Some species move into shade when the ambient temperature starts increasing.
III. Some species burrow into the soil and escape from the above ground heat.
The above statements depict the adaptations in
(a) kangaroo rat (b) desert lizards
(c) mammals (d) humans

114. Given below are some characteristic features.
I. A ventral heart and closed circulatory system is present.
II. Haemoglobin is present in red blood corpuscles.
III. Digestive system is well-developed
IV. Hepatic portal system is present.
Identify the phylum in which above characteristic features are included
(a) Chordata (b) Non-Chordata
(c) Porifera (d) Hemichordata

115. Which statement is incorrect regarding fermentation?
I. It is anaerobic respiration which does not require oxygen.
II. It generally utilises NADH produced during glycolysis.
III. It does not produce ATP.
IV. It produces pyruvic acid.
Choose the correct option.
(a) I and IV (b) II, III
(c) Only IV (d) Only II

116. Identify the incorrectly matched pair.
(a) Codominance — ABO blood grouping in humans
(b) Pleiotropy — Phenylketonuria
(c) Polygenic inheritance — Human skin colour
(d) Incomplete dominance — Coat colour in mice

117. *Adiantum caudatum* is also called walking fern because
(a) it's leaves are in the shape of limbs
(b) new plantlets grow wherever leaves touch the ground
(c) the petiole is shaped like a leg
(d) the plant shifts from one place to another sometimes

118. Given below are statements regarding viruses. Identify the correct statement.
(a) All viruses contain both RNA and DNA
(b) They are obligate parasites
(c) Viruses possess their own metabolic system
(d) Viral nucleic acid is called as capsid

119. Manganese becomes toxic when absorbed by plants in higher amounts. It is due to
(a) reduction in uptake of Fe^{3+} and Mg^{2+}
(b) inhibition of binding of Mg^{2+} to specific enzymes
(c) inhibition of Ca^{2+} translocation in shoot apex
(d) All of the above

120. One of the following statements is incorrect with reference to mycoplasma. Identify it.
(a) Absence of cell wall
(b) Covered with three layered plasma membrane
(c) 70S ribosomes are present
(d) They are Gram positive

121. Which of the following is the correct sequence of electron carriers between PS-II to PS-I in photosynthesis?
(a) PQ, *cyt*-b_6, *cyt*-f, PC (b) *cyt*-b_6, *cyt*-f, PC, PQ
(c) PC, *cyt*-b_6, *cyt*-f, PQ (d) *cyt*-f, *cyt*-b_6, PQ, PC

122. Which option shows the correctly matched hormone with its source and function?

	Hormone	Source	Function
(a)	MSH	Pars intermedia	Darkening of skin
(b)	STH	Pars nervosa	Micturition
(c)	ACTH	Pars tuberalis	Control's blood glucose level
(d)	FSH	Pars distalis	Cushing's disease

123. Which among the following is an example of root climbers?

(a) *Pothos* (b) *Cuscuta*
(c) *Linospora* (d) *Bauhinia*

124. Find the odd one out in the following list.

I. *Pila* II. *Aplypsia*
III. *Unio* IV. *Octopus*
V. *Doris* VI. *Limax*

Choose the correct option.

(a) Only I (b) I and VI (c) III and IV (d) Only IV

125. In non-cyclic photophosphorylation, which of the following electron carriers is a copper containing compound?

(a) Ferredoxin (b) Cytochrome-a_3
(c) Plastocyanin (d) Plastoquinone

126. Identify the incorrectly matched pair regarding the development of plasmid vectors in following three phases.

(a) First phase → Poor replication ability
(b) Second phase → With unsuitable selectable markers
(c) Third phase → Incorporation of high expression promoters for expression of large amounts of foreign proteins
(d) None of the above

127. Which among the following statements is/are correct regarding active absorption?

(a) ATP is required
(b) Rate of respiration increases during absorption
(c) Both (a) and (b)
(d) None of the above

128. Which among the following statements is/are correct regarding dicot leaf ?

(a) Leaves are dorsiventral
(b) Bulliform cells are absent
(c) Vascular bundle is surrounded by parenchymatous bundle sheath
(d) All of the above

129. Identify the correctly matched pairs.

I. RNA polymerase I → Transcribes *r*RNAs
II. RNA polymerase II → Transcribes precursor of *m*RNA
III. RNA polymerase III → Transcribes *t*RNA

The correct option with matched pairs is

(a) I and II (b) II and III
(c) I, II and III (d) I and III

130. Match Column I with Column II and choose the correct option from the codes given below.

	Column I (Gene transfer)		Column II (Characteristics)
A.	Plant tumours	1.	Delivering large genes into bacterial cells
B.	Bacteriophages	2.	Delivering genes to living patients in gene therapy
C.	Adenoviruses	3.	Foreign genes are replicated into every daughter cell
D.	Retroviruses	4.	Usage of *Agrobacterium tumefaciens*

Codes

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	4	1	2	3
(d)	1	2	4	3

131. Which of the following statements are not true for the synergids?

I. They are generally two in number and occur laterally to egg cell.
II. These are characterised by the presence of finger-like filiform apparatus towards chalazal pole.
III. They secrete chemotactic substances to keep the egg cell safe.
IV. They have polarised protoplasm where nuclei occur towards the micropylar pole and vacuoles towards the central cell.
V. Synergids degenerate before the entry of the pollen tube into the embryo sac.

Choose the correct option.

(a) II and III (b) II, IV and V
(c) I, IV and V (d) All of these

132. Tubular reabsorption involves

I. absorption of water and useful solutes from the glomerular filtrate into blood.
II. It occurs mostly in PCT.
III. It takes place only by active transport.
IV. It occurs mostly in DCT.

The correct statements are

(a) I and II (b) II and III (c) III and IV (d) I and IV

133. The top floor of the building caught fire due to short circuit. The scared residents of that floor immediately started running downstairs. Which hormone instigated this action?

(a) Thyroxine (b) Calcitonin
(c) Somatotropin (d) Adrenaline

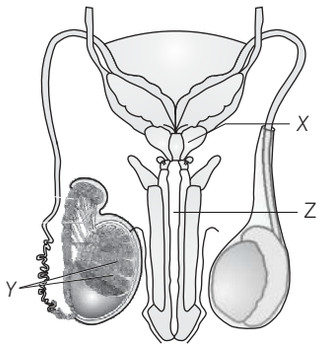
134. Match Column I with Column II and choose the correct option from the codes given below.

	Column I (Terms used in ecology)		Column II (Description)
A.	Community	1.	All organisms that live in a place together with their physical environment
B.	Ecosystem	2.	group of ecosystems that share similar climates
C.	Biome	3.	entire planet with all its organisms and physical environments
D.	Biosphere	4.	assemblage of different populations

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	1	4	2	3
(c)	1	4	3	2	(d)	4	1	2	3

135. Given below is the diagrammatic presentation of a male reproductive tract.



Which of the following correctly identifies part *X, Y* and *Z* along with their functions?

(a) *X* – Prostate gland, Function – Secrete testosterone hormone
(b) *X* – Urethra, Function – Outlet for urine and semen
(c) *Y* – Testis, Function – Primary male reproductive organs
(d) *Z* – Seminal vesicle, Function – Secretes sperms

136. Identify the incorrectly matched pair in reference to sewage treatment.

(a) Primary treatment → Physical removal of particles
(b) Secondary treatment → BOD is reduced
(c) Flocs → Masses of bacteria associated with fungal filaments
(d) Greater BOD → Less is the polluting potential

137. Which among the following statements is correct in reference to promoter of transcription unit?

I. It is located towards 5′ end (upstream) of the structural gene.
II. It provides the site for binding of RNA polymerase.
III. The template and coding strands of DNA are defined by the promoter site.
IV. The site is A and T rich region called TATA BOX (in prokaryotes) and Pribnow Box (in eukaryotes).

Choose the correct option.

(a) I and II (b) III and IV
(c) I, III and IV (d) All of these

138. Identify the incorrectly matched pair.

(a) Pink pages — Critically endangered species
(b) Green pages — Species which are no longer threatened
(c) Both (a) and (b)
(d) None of the above

139. Arrange the following in sequence of steps in plant breeding.

I. Collection of variability.
II. Cross hybridisation among selected parents.
III. Evaluation and selection of parents.
IV. Selection and testing of superior recombinants.
V. Testing, release and commercialisation of new cultivars.

The correct sequence of steps in plant breeding is as follows.

(a) I, II, III, IV, V (b) I, III, II, IV, V
(c) I, II, IV, V, III (d) I, II, III, V, IV

140. *E. histolytica*, *E. coli* and *E. gingivalis* are a group of

(a) symbiotic bacteria present in the intestine of most primates
(b) protozoan parasites found in gastrointestinal tract
(c) bacterial parasites that can be dealt with immunisation at proper time
(d) None of the above

141. Which among the following statements is incorrect regarding a food web?

(a) It consists of number of interconnected food chains
(b) It is never straight
(c) A number of alternate sources of food are present
(d) None of the above

142. Identify the correct plant growth regulator(s) for the formation of the corresponding plant tissue?

	Plant Growth Regulator	Plant Tissue
(a)	Auxin	Shoots
(b)	Auxin and cytokinin	Callus
(c)	Abscisic acid	Stem elongation
(d)	Cytokinin	Somatic embryo

143. Which among the following is/are correctly matched regarding age pyramid?
(a) Young population pyramid — Pyramid with broad base
(b) Stable population — Bell-shaped pyramid
(c) Declining population — Urn-shaped pyramid
(d) All of the above

144. Identify the incorrectly matched pair.
(a) Biopiracy — Exploit biological resources of other nations
(b) Biopatent — Right granted by the government to an inventor to prevent others to make commercial use of one's invention
(c) Both (a) and (b)
(d) None of the above

145. The functions of endosperm include
(a) Forms sufficient food for the developing embryo
(b) Stores carbohydrates, fats, proteins, etc
(c) Its contents are necessary for the growth of embryo
(d) All of the above

146. The secondary immune response is not characterised by
(a) more rapid
(b) lasts for longer periods
(c) occurs at the second and subsequent exposure of the same host to same antigen
(d) takes longer time to establish immunity

147. Which among the following statements is/are correct regarding phagemids?
(a) They give high yield of foreign DNA
(b) They can provide single-stranded DNA
(c) Due to their small size single-stranded DNA upto 10 kbp can be easily obtained from them
(d) All of the above

148. During expiration,
I. external intercostal muscles relax.
II. diaphragm bends upwards.
III. diaphragm straightens.
IV. external intercostal muscles contract.
The correct statements are
(a) I and III (b) I and II
(c) III and IV (d) I and IV

149. Resource partitioning is
(a) avoidance of competition
(b) realised niches divide resources among several species
(c) Both (a) and (b)
(d) None of the above

150. The reason why the skeletal muscles are syncytial while smooth and cardiac muscles are not, is
(a) skeletal muscle is made up of a number of muscle bundles held together
(b) quantity of myoglobin is high
(c) smooth muscles and cardiac muscles are more active than skeletal muscles
(d) Both (b) and (c) explain it correctly

151. Eukaryotic mature *m*RNAs have a 5′ cap that is residue of
(a) adenylic acid (b) adenine
(c) 3-methyl guanosine (d) 7- methyl guanosine

152. Which one of the following statements is correct for Okazaki fragments ?
(a) They are DNA - RNA hybrids
(b) They are double-stranded
(c) They are formed due to nicking of the sugar-phosphate back bone of parental strand
(d) They contain covalently linked RNA and DNA

153. The properties of genetic material include
(a) it should be able to replicate itself
(b) it should be stable both chemically and structurally
(c) it should be able to express itself in the form of 'Mendelian characters'
(d) All of the above

154. Lethal genes are
(a) responsible for reappearance of ancestral traits
(b) always recessive
(c) present on separate chromosomes and influence the same trait
(d) lethal and cause death of the organism in homozygous state

155. Gauri visited her village with her parents for the first time. She was surprised to see many infants in her village who were very weak with stunted growth, slender legs and protruding belly. Which one of the following diseases are they likely to be suffering from?
(a) Kwashiorkor (b) Marasmus
(c) Pellagra (d) Xerophthalmia

156. Identify the incorrectly matched pair.
(a) Dicot root — Pith absent or poorly developed
(b) Monocot root — Cambium absent
(c) Both (a) and (b)
(d) None of the above

157. Read the following statements.
I. These muscle fibres are supplied with both central and autonomic nervous system.
II. They are not under the will of the animal.
III. These fibres never get fatigue.
In reference to above statements, which of the following muscles possess the above characteristics?
(a) Smooth muscles (b) Cardiac muscles
(c) Striated muscles (d) None of these

158. Which part acts as a highway for the relay of many signals to and from the cerebrum and cerebellum?
(a) Pons Varolii (b) Medulla oblongata
(c) Cerebellum (d) Corpora quadrigemina

159. The four kingdom system of classification consists of
(a) Monera, Protista, Plantae, Animalia
(b) Monera, Protista, Fungi, Plantae
(c) Monera, Protista, Fungi, Animalia
(d) Monera, Protista, Fungi, Virus

160. Arrange the following in reference to pollen development in flowering plants.
I. Microspore mother cell develops within pollen sac.
II. Each microspore undergoes mitosis to form immature pollen grain.
III. Microspore mother cell undergoes meiosis to form four haploid microspores.
IV. Generative cell undergoes mitosis to form two sperm cells.
Choose the correct option.
(a) I → II → III → IV (b) I → III → II → IV
(c) III → I → II → IV (d) IV → I → II → III

161. Identify the incorrect statement.
(a) In Arthropoda, jointed appendages help in locomotion
(b) Organs of Bojanus are present in Mollusca
(c) Ambulacral system is present in Echinodermata
(d) Roundworm is an example of Platyhelminthes

162. Identify the incorrectly matched pair.
(a) Siliqua → *Capsella*
(b) Silicula → *Iberis*
(c) Capsule → Cotton
(d) Follicle → *Calotropis*

163. Which statement is incorrect regarding *lac* operon?
(a) Lactose is known to be the inducer
(b) *z*-gene codes for β - galactosidase
(c) *y*-gene codes for permease
(d) *a*-gene codes for the repressor

164. What may be caused as a result of a marriage between close relatives?
(a) Erythroblastosis foetalis
(b) Mutations
(c) More recessive alleles will come together
(d) Incomplete dominance

165. Which of the following is correctly matched pair?
(a) Leptotene — Homologous chromosomes get associated as bivalents
(b) Zygotene — Chromatin begins to get compacted showing fine threads
(c) Pachytene — Homologous chromosomes come in pairs
(d) Diplotene — Chiasma formation and separation of chromosomes

166. Invertebrates have a bluish blood because of respiratory pigment
(a) haemoglobin (b) haemocyanin
(c) cytochrome (d) chlorocruorin

167. In dorsiventral leaf
(a) guard cells are kidney-shaped
(b) mesophyll is differentiated into palisade and spongy parenchyma
(c) bundle sheath cells are colourless
(d) bulliform cells (motor cells) absent

168. It two genes are located very close to each other on the same chromosome
(a) no linkage is there
(b) crossing over increases
(c) occurrence of crossing over is very rare
(d) they merge to form a larger gene

169. The bulk of the semen is contributed by
(a) prostate gland secretion (b) seminal vesicle fluid
(c) bulbourethral gland's fluid (d) Cowper's gland secretion

170. Which among the following statements is/are correct regarding Protista?
(a) Unicellular eukaryotes
(b) Possess double nuclear membrane
(c) They are the smallest free-living microorganisms
(d) Both (a) and (b)

171. Intermediate host is
I. in which asexual reproduction occurs.
II. in which the immature form harbors.
III. in which sexual reproduction occurs.
IV. which harbors the mature form.
The option with correct statements is
(a) I and II (b) III and IV (c) III and II (d) I and IV

172. Which among the following statements is/are correct regarding bryophytes?
(a) They are non-vascular cryptogams
(b) Sporophyte is attached to gametophyte
(c) Gametophyte is the dominant phase in life cycle
(d) All of the above

173. Match the items given in Column I with these in Column II and select the correct option.

	Column I (Plant hormones)		**Column II** (Action)
A.	IAA	1.	Stomatal closure
B.	Gibberellins	2.	Stem elongation
C.	Cytokinins	3.	Cell wall elongation
D.	Abscisic acid	4.	Cell division

Codes

	A	B	C	D
(a)	3	2	4	1
(c)	4	3	2	1

	A	B	C	D
(b)	1	2	3	4
(d)	2	4	1	3

174. Identify the incorrect pair.

(a) G_0–Non-dividing
(b) G_1–Growth and differentiation
(c) G_2–Chromosomes are replicated
(d) S–Synthesis of DNA

175. Which of the following male sex organs are also known as bulbouretrhal glands?

(a) Prostate glands (b) Pineal glands
(c) Bartholin glands (d) Cowper's glands

176. Match Column I with Column II and choose the correct option from the codes given below.

	Column I (Type of pollutant)		Column II (Description)
A.	Primary pollutant	1.	Present in environment as it is produced
B.	Secondary pollutant	2.	Formed from primary pollutant
C.	Qualitative pollutant	3.	Produced by anthropogenic activities
D.	Quantitative pollutant	4.	Concentration reaches more than threshold value

Codes

	A	B	C	D
(a)	1	2	3	4
(b)	1	2	4	3
(c)	2	1	3	4
(d)	2	1	4	3

177. Deuterostomes are the ones in which

I. anus develops first. II. mouth develops later.
III. mouth develops first. IV. anus develops later.

Choose the correct option.

(a) I and II (b) II and IV
(c) I and III (d) None of these

178. Match Column I with Column II and choose the correct option from the codes given below.

	Column I (Modifications of stem)		Column II (Characteristic features)
A.	Prickles	1.	Bear several nodes and internodes
B.	Phylloclade	2.	One internode long
C.	Cladodes	3.	Help in vegetative propagation
D.	Bulbils	4.	Act as climbing organs

Codes

	A	B	C	D
(a)	4	1	2	3
(b)	1	2	3	4
(c)	4	1	3	2
(d)	1	4	3	2

179. Which statement(s) is/are correctly related to *in situ* conservation?

(a) It involves protection of species threatened with extinction
(b) Flora and fauna exist, without human interference
(c) It involves protection of species outside their natural homes
(d) Both statements (a) and (b)

180. Which assumptions should be met in Hardy-Weinberg equilibrium?

(a) The population size is very large
(b) Individuals pair by chance, not according to their genotypes or phenotypes
(c) no gene flow
(d) All of the above

